The Quanto Adjustment and the Smile

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Abstract

Quanto derivatives allow investors of a given country to invest directly in assets of countries with different currencies without taking any exposure to the evolution of the exchange rate. European quanto derivatives are usually priced using the well known quanto adjustment corresponding to the forward of the quantoed asset under the assumptions of the Black-Scholes model. In this article, I present the quanto adjustment corresponding to the local volatility model that allows pricing quanto derivatives consistently with the observed market equity skew and exchange rate smile. I then examine the model risk arising in the standard quanto adjustment by fitting the local volatility model to market data and then comparing the prices of European quanto euro derivatives on the Nikkei 225 index with those generated by the standard quanto adjustment. The results show that the standard quanto adjustment can be subject to significant pricing errors when compared with the local volatility model. I also compare the pricing performance of the local volatility model with a multivariate stochastic volatility model. The results show that when the correlation between the instantaneous variances associated with the underlying asset and the exchange rate is close to one, as it is the case when we consider historical data, there is little evidence of model risk for the local volatility model in the pricing of European quanto euro derivatives on the Nikkei 225 index.

Keywords: quanto adjustment, implied volatility skew, implied volatility smile, local volatility, stochastic volatility, instantaneous correlation, model risk.

JEL: G12, G13.

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1 Introduction

Quanto options are one type of multi-asset derivatives that allow investors of a given country to invest in assets of countries with another currency without assuming exposure to the exchange rate. One example could be a European call on the Nikkei 225 index, where the positive performance of the index with respect to the strike price is paid directly in euros. This contract is called European quanto euro call on the index. As Derman (2004) points out, the quanto option, which started in the early 1990s, was one of the first exotic structures and it became quite popular because its payoff reflected investors' needs.

Under the assumptions of the Black-Scholes (1973) model it is well known that it is possible to introduce a so-called quanto correction in the forward of the underlying asset to price quanto derivatives. The derivation of this quanto adjustment can be found, for example, in Derman et al. (1990), Reiner (1991), Wilmott (2006) or Hull (2006).

Option prices are usually quoted using implied volatilities obtained from the Black-Scholes (1973) option pricing formula. The implied volatility expressed as a function of the strike and the maturity is known as the time t implied volatility surface. The assumptions of the Black-Scholes (1973) model imply that the implied volatility surface should be flat and static. But since the stock market crash on October 1987, equity options markets have been characterized by a persistent negative dependence of the implied volatility with respect to the strike price, which is known as the implied volatility skew. This stylized fact has been documented by Derman and Kani (1994), Dupire (1994), Rubinstein (1994) or Dumas et al. (1998). In currency markets out-of-the-money puts, as well as out-of-the-money calls usually exhibit higher implied volatilities than at-the-money options. This fact is commonly referred to as the volatility smile and has been documented by Rebonato (1998), Jex et al. (1999) or Castagna and Mercurio (2007).

In the literature there are a number of models that accounts for the existence of volatility skew and/or smile. Namely stochastic volatility, jumps models, local volatility and variance gamma models. Stochastic volatility models postulate that volatility follows a stochastic process possibly correlated with the process for the stock price. Examples of this approach can be found in the works of Hull and White (1987) and Heston (1993). Merton (1976), among others, incorporates the possibility of jumps in the stochastic process for the underlying asset price that allows for the existence of volatility skew. Local volatility models postulate that the instantaneous volatility (called local volatility) is a deterministic function of the underlying asset price and time. Different approaches to obtain the local volatility from the market price of European options can be found in Dupire (1994), Derman and Kani (1994), Andersen and Brotherton-Ratcliffe (1998) or Brown and Randall (1999). Finally, the variance gamma process for the stock price was introduced by Madan et al. (1998). This process generalizes the Brownian Motion and allows for the existence of skewness and kurtosis in the return distribution.

Jäckel (2010a) prices quanto derivatives using the displaced diffusion model introduced by Rubinstein (1983). Unfortunately, this model allows for the underlying asset to attain negative values and this circumstance can affect the pricing results. On the other hand, Jäckel (2010b) offers examples of the pricing of quanto derivatives under a stochastic volatility model derived by Jäckel and Kalh (2008) and under the common quanto adjustment. Jäckel (2010b) shows numerical examples of price discrepancies between the two models for long-term quanto options. Although this results are quite interesting, unfortunately, the implied volatility surface associated with the model introduced by Jäckel and Kalh (2008) is not consistent with the no-arbitrage conditions derived by Lee (2004) for the asymptotic value of implied volatilities.

As Madan et al. (2007) points out, the local volatility model usually captures the surface of implied volatilities more precisely than other approaches, such as stochastic volatility models. Hence, in this article I consider the pricing of quanto derivatives in the presence of equity volatility skew and currency volatility smile using the local volatility approach. I derive the stochastic differential equation corresponding to the quantoed underlying asset and I compare it with the usual quanto adjustment under the Black-Scholes (1973) model assumptions to investigate the importance of the equity skew and the currency smile in the correct valuation of quanto options. To my knowledge, a study on smile-consistent valuation of quanto derivatives using real market data has not been published yet. The objective of this article is to fill this gap. To this end, I use market data to price European quanto euro options on the Nikkei 225 index. This type of options has become very popular between European investors who want to participate in the performance of the Nikkei 225 index without any exposure to the evolution of the foreign exchange rate. The empirical results show that, although it could seem that quanto options are quite standard, the usual quanto correction is subject to significant pricing errors when compared with the local volatility model. Importantly, the pricing errors are higher for long-term maturities and depend on the parameters values, such as the level of correlation between the underlying asset and the exchange rate. Therefore, the pricing of quanto options can be quite model dependent and, in this sense, can be exposed to model risk which is the risk arising from the use of an inadequate model.

The growth experienced in recent years in both the variety and volume of structured products, which are trade in the over-the-counter market, implies that banks and other financial institutions have become increasingly exposed to model risk. In fact, as Derman and Wilmott (2009) pointed out, one of the reasons of the financial crisis that started in August 2007 with the collapse in subprime mortgages was the use of inadequate models to valuate complex credit derivatives. In this sense, the Basel Committee on Banking Supervision[1] has recently stated that it will supplement the risk-based capital requirements with a leverage ratio that intends to introduce additional safeguards against model risk and measurement error.

Some researchers, such as Dumas et al. (1998), have tested the effectiveness of the local volatility model when it is calibrated to market data at one point in time and used to price options at a later time. As Hull and Sou (2002) points out, although this approach is interesting, unfortunately it does not capture the essence of the risk in the models as they are used in trading rooms. Traders typically calibrate the model to plain vanilla instruments and use it to price exotic options at the same time. The performance of the model pricing vanilla options at a later time is not necessarily an indication of its suitability to price exotic options. Hull and Sou (2002) did not find evidence of model risk for the local volatility model in the pricing of European style options, such as compound options. This fact indicates that the local volatility model can be a correct approach to price European quanto derivatives in the presence of volatility smile and/or skew. However, since the local volatility is a deterministic function of time and the spot price, it is well known that the local volatility model does not properly accounts for the existence of volatility in the volatility. To analyze the impact of this effect in the pricing of quanto derivatives, I also compare the pricing performance of the local volatility model with the prices corresponding to quanto derivatives under a multivariate version of the Heston (1993) stochastic volatility model. This is an important contribution since, to my knowledge, it represents the first analysis of the pricing discrepancies associated with the local volatility model and with a stochastic volatility framework in the valuation of quanto derivatives. Moreover, as Wilmott (2006) points out, the local volatility and stochastic volatility models are the most widely used models by financial institutions to price exotic options.

The rest of the article proceeds as follows. Section 2 presents the usual quanto adjustment under the Black-Scholes (1973) assumptions, as well as the quanto correction corresponding to

[1]See Basel Committee on Banking Supervision (2010).

the local volatility model. This section also analyzes the price sensitivities of European quanto derivatives and it offers insight regarding the correct hedging of the risks arising from these derivatives. Section 3 compares the models presented in section 2 in the pricing of European quanto euro options on the Nikkei 225 index. Section 4 analyzes the pricing discrepancies corresponding to the local volatility model and the multivariate version of the Heston (1993) model in the pricing of quanto derivatives. Finally, section 5 offers concluding remarks.

2 The quanto adjustment in the presence of skew and smile

2.1 Quanto adjustment in a local volatility framework

Let $S(t)$ denote the spot price of the underlying asset S at time $t \in [0, \Upsilon]$, where Υ is some arbitrarily distant horizon. Let assume that S is observed in currency Y. Consider a quanto derivative that provides a payoff in currency X at time $t = T$. For simplicity, I assume that the continuously compounded risk-free rates in both currencies r_Y and r_X, as well as the dividend yield q are constant. Let $P_J(t, T)$ (for $J = X, Y$) denote the time t price of a zero coupon bond which pays one unit of currency J at time $t = T$.

Let define the foreign exchange rate $Z = \frac{Y}{X}$ as the number of units of currency Y per unit of currency X. Finally, consider the probability measure Q_Y, defined on a probability space $(\Omega, \mathcal{F}, Q_Y)$ such that asset prices expressed in terms of the current account in currency Y are martingales. We denote this probability measure as the risk-neutral measure with respect to currency Y. Analogously, Q_X is the risk-neutral probability measure with respect to currency X.

I assume that the risk-neutral assets prices evolutions under Q_Y are governed by the following stochastic differential equations:

$$\begin{aligned}
\frac{dS(t)}{S(t)} &= (r_Y - q)\, dt + \sigma_S(t, S)\, dW_S^{Q_Y}(t) \\
\frac{dZ(t)}{Z(t)} &= (r_Y - r_X)\, dt + \sigma_Z(t, Z)\, dW_Z^{Q_Y}(t)
\end{aligned}$$

where $W_S^{Q_Y}$ and $W_Z^{Q_Y}$ are Wiener processes under Q^Y such that:

$$dW_S^{Q_Y}(t)\, dW_Z^{Q_Y}(t) = \rho_{SZ} dt \tag{1}$$

with ρ_{SZ} being the instantaneous correlation between the underlying asset and the exchange rate, which is assumed to be constant. The term $\sigma_S(t, S)$ is the local volatility corresponding to the underlying asset and it is a deterministic function of time and the spot price. Analogously, $\sigma_Z(t, Z)$ represents the local volatility associated with the exchange rate.

Consider an arbitrary asset V denominated in currency Y. From the Fundamental Theorem of Asset Pricing we have that:

$$\frac{V(0)}{P_Y(0,t)} = E_{Q_Y}[V(t)]$$

On the other hand, it is also possible to express the price of this asset in terms of the risk-neutral probability measure associated with currency X:

$$\frac{V(0)}{Z(0)\,P_X(0,t)} = E_{Q_X}\left[\frac{V(t)}{Z(t)}\right]$$

Combining the previous equations gives the following relationship between both probability measures:

$$E_{Q_X}[M(t)] = E_{Q_Y}\left[M(t)\,\frac{Z(t)}{Z(0)}\frac{P_Y(0,t)}{P_X(0,t)}\right]$$

where $M(t) = \frac{V(t)}{Z(t)}$ and where

$$\frac{dQ_X}{dQ_Y} = \frac{Z(t)}{Z(0)}\frac{P_Y(0,t)}{P_X(0,t)}$$

is the Radon-Nikodym derivative.

Let $F_Z^{Q_Y}(t,T)$ denote the time t price of the forward contract on the exchange rate with maturity $t = T$, under the probability measure Q_Y, which verifies that:

$$\frac{dF_Z^{Q_Y}(t,T)}{F_Z^{Q_Y}(t,T)} = \sigma_Z(t,Z)\,dW_Z^{Q_Y}(t)$$

Applying Ito's lemma to the previous expression and integrating the result over the interval $[0,T]$ gives:

$$F_Z^{Q_Y}(T,T) \equiv Z(T) = F_Z^{Q_Y}(0,T)\,e^{-\frac{1}{2}\int\limits_0^T \sigma_Z^2(t,Z)dt + \int\limits_0^T \sigma_Z(t,Z)dW_Z^{Q_Y}(t)} \tag{2}$$

On the other hand, covered interest rate parity implies that:

$$F_Z^{Q_Y}(0,T) = Z(0)\,\frac{P_X(0,T)}{P_Y(0,T)} \tag{3}$$

Combining equations (2) and (3) yields the following expression for the Radon-Nikodym derivative:

$$\frac{dQ_X}{dQ_Y} = e^{-\frac{1}{2}\int\limits_0^t \sigma_Z^2(u,Z)du + \int\limits_0^t \sigma_Z(u,Z)dW_Z^{Q_Y}(u)}$$

Hence, from the Girsanov theorem[2], the process $W_Z^{Q_X}(t)$ defined by:

$$dW_Z^{Q_X}(t) = dW_Z^{Q_Y}(t) - \sigma_Z(t, Z)\, dt$$

is a Wiener process under the probability measure Q_X. Therefore, we have the following stochastic differential equation corresponding to the evolution of the exchange rate under the probability measure Q_X:

$$\frac{dZ(t)}{Z(t)} = \left[r_Y - r_X + \sigma_Z^2(t, Z)\right] dt + \sigma_Z(t, Z)\, dW_Z^{Q_X}(t) \tag{4}$$

On the other hand, taking into account equation (1), it is possible to express the law of behavior for the quantoed stock under the probability measure Q_X as follows[3]:

$$\frac{dS(t)}{S(t)} = \left[r_Y - q + \rho_{SZ}\sigma_Z(t, Z)\,\sigma_S(t, S)\right] dt + \sigma_S(t, S)\, dW_S^{Q_X}(t) \tag{5}$$

where the process $W_S^{Q_X}(t)$ defined by:

$$dW_S^{Q_X}(t) = dW_S^{Q_Y}(t) - \rho_{SZ}\sigma_Z(t, Z)\, dt$$

is a Wiener process under the probability measure Q_X and where:

$$dW_S^{Q_X}(t)\, dW_Z^{Q_X}(t) = \rho_{SZ} dt$$

Note that the existence of a local volatility function in the exchange rate and in the underlying asset transfers into a localized drift term in the evolution of the underlying asset, as well as in the evolution of the exchange rate, under the probability measure Q_X. These results indicate the important effects of the smile and the skew on the dynamics of the quantoed assets.

2.2 Quanto adjustment under constant instantaneous volatilities

Let assume that we are in a Black-Scholes (1973) framework, so that $\sigma_Z(t, Z) = \sigma_Z$ and $\sigma_S(t, S) = \sigma_S$ are constant. In this case, applying Ito's lemma to equation (5), integrating the result over the interval $[0, t]$ and taking the expectation with respect to the probability measure Q_X yields the well known adjustment corresponding to the forward of the quantoed underlying

[2]See for example Musiela and Rutkowski (1998).

[3]Madan et al. (2007) derive a similar expression to the one of equation (5) applying the Gyöngy (1986) result, who showed how to associate with a general Ito process a Markov process with the same marginal distribution.

asset:

$$F_S^{Q_X}(0,t) = F_S^{Q_Y}(0,t)\, e^{\rho_{SZ}\sigma_S\sigma_Z t} \tag{6}$$

where $F_S^{Q_X}(0,t) = E_{Q_X}[S(t)]$ and where:

$$F_S^{Q_Y}(0,t) = E_{Q_Y}[S(t)] = S(0)\, e^{(r_Y - q)t}$$

is the forward contract on the underlying asset under the probability measure Q_Y. As pointed out by Castagna and Mercurio (2007), since in financial markets implied volatilities vary randomly through time, many forex option traders run their books by revaluing and hedging according to the Black-Scholes (1973) model, with the at-the-money implied volatility being frequently updated to match the actual market level. In this case, equation (6) has important implications for the risk management of quanto derivatives. Consider a financial institution that is long the quanto forward. The vega with respect to the exchange rate volatility is given by:

$$\frac{\partial F_S^{Q_X}(0,t)}{\partial \sigma_Z} = F_S^{Q_X}(0,t)\, \rho_{SZ}\sigma_S t$$

The previous equation shows that the vega of the quanto forward with respect to the exchange rate volatility is not directly linked to a concrete level of the exchange rate but it is a function of the stock price and the model parameters. This fact implies that the quanto forward generates an exposure to the exchange rate volatility over all the range of values associated with the exchange rate. Therefore, it is not enough to trade an at the-money option on the exchange rate to hedge this vega risk. By contrast, the financial institution would need a continuum of options with different strike prices and the vega required on each strike would be a function of the underlying asset price and the model parameters. Analogously, the quanto forward exhibits sensitivity with respect to the underlying asset volatility for different levels of the stock price and this sensitivity is higher, in absolute terms, the higher the value of the underlying asset. In this sense, the quanto forward is somewhat similar to a gamma swap, which is a forward contract on the variance of a given asset, where this variance is weighted by the price of the underlying asset[4].

[4]Formally, in continuous time the payoff at maturity of a gamma swap is given by:

$$\frac{1}{T}\int_0^T S_t \sigma_t^2 dt - K_{gs}$$

where T is the maturity, σ_t^2 is the realized variance and is K_{gs} the strike price of the swap or gamma swap rate.

On the other hand, the sensitivity of the quanto forward price with respect to the instantaneous correlation is given by:

$$\frac{\partial F_S^{Q_X}(0,t)}{\partial \rho_{SZ}} = F_S^{Q_X}(0,t)\,\sigma_Z \sigma_S t$$

This sensitivity is higher, the higher the value of the forward $F_S^{Q_X}(0,t)$, the longer the maturity and the higher the volatilities σ_Z and σ_S. As an example, let us assume that $r_Y = 0.50\%$, $r_X = 1.50\%$, $q = 2\%$, $\rho_{SZ} = 0.75$, $\sigma_S = 25\%$ and $\sigma_Z = 18\%$. In this case, the sensitivity of a quanto forward with maturity equal to three years with respect to a 0.01 change in the instantaneous correlation ρ_{SZ} is equal to 0.14%.

The previous examples show that, even for the simple quanto adjustment under the Black-Scholes (1973) model, the correct hedge of the risks associated with the quanto derivatives can be quite complex to implement. The examples also indicate that the consideration of the market equity skew and exchange rate smile is fundamental for a correct valuation of quanto derivatives, given that they exhibit sensitivity to volatility over different ranges of prices corresponding to the underlying asset and the exchange rate.

Let $C_{Q_X}(0,T,K)$ denote the time $t=0$ price of a European quanto call with maturity $t=T$ and strike price K, which pays at expiration the quantity $(S_T - K)^+$ denominated in units of currency X. The price of this option under the assumptions of the Black-Scholes (1973) model is:

$$C_{Q_X}(0,T,K) = P_X(0,T)\left[F_S^{Q_X}(0,T)\,\Phi(d) - K\Phi\left(d - \sigma_S\sqrt{T}\right)\right] \tag{7}$$

where $\Phi(.)$ is the standard normal cumulative distribution function and where:

$$d = \frac{1}{\sigma_S\sqrt{T}}\left[\ln\left(\frac{F_S^{Q_X}(0,T)}{K}\right) + \frac{\sigma_S^2}{2}T\right]$$

From the put-call parity, the price of the corresponding quanto put $P_{Q_X}(0,T,K)$ is given by:

$$P_{Q_X}(0,T,K) = P_X(0,T)\left[K\Phi\left(\sigma_S\sqrt{T} - d\right) - F_S^{Q_X}(0,T)\,\Phi(-d)\right] \tag{8}$$

The above equation shows that in general, unlike what happens with standard options, the vega of the quanto put with respect to the volatility of the asset and the exchange rate, is different from the vega of the call with the same strike price, given that the quanto forward exhibit sensitivity to both volatilities. In particular, the vega of the call option with respect to the volatility of the stock is:

$$\frac{\partial C_{Q_X}(0,T,K)}{\partial \sigma_S} = P_X(0,T)\, F_S^{Q_X}(0,T)\left[\sqrt{T}\phi(d) + \sigma_Z \rho_{SZ} T \Phi(d)\right]$$

where $\phi(.)$ is the standard normal density function, whereas we have the following expression for the vega of the quanto put with respect to σ_S:

$$\frac{\partial P_{Q_X}(0,T,K)}{\partial \sigma_S} = \frac{\partial C_{Q_X}(0,T,K)}{\partial \sigma_S} - P_X(0,T)\, F_S^{Q_X}(0,T)\,\sigma_Z \rho_{SZ} T$$

Hence, the vega of the put will be lower than the vega of the call for $\rho_{SZ} > 0$. Conversely, the vega of the put will be greater than the vega of the call for $\rho_{SZ} < 0$. As it will be shown in the next section, different levels for the instantaneous correlation between the exchange rate and the underlying asset, which change the vega sensitivity associated with puts and calls, can lead to important pricing discrepancies corresponding to quanto derivatives under the local volatility model and under the assumption of flat implied volatilities.

3 Local volatility versus flat implied volatilities

In this section I compare the performance of the standard quanto adjustment under the Black-Scholes (1973) assumptions and the quanto adjustment corresponding to the local volatility model, in the pricing of European quanto euro derivatives on the Nikkei 225 index. There are two major reasons for this choice. The first one is that these products have become quite popular between euro-based investors and there are a lot of structured products that include this kind of options. The second one has to do with the level of correlation between the Nikkei 225 index and the euro-yen exchange rate. Figure 1 shows the realized correlation between the weekly performances[5] of these variables obtained for Bloomberg. We can see from the figure that the correlation jumped roughly from a level of 0.40 to almost 0.80 during the period August 2007 to December 2007, with the start of the international financial crisis, and it has remained high since then. Note that the greater the correlation, in absolute terms, the greater the magnitude of the quanto adjustment. In this case, there are greater potential pricing discrepancies between different models.

To accomplish the study, I follow an argument that is similar to the one used by Hull and

[5]I selected weekly data to avoid asynchrony in the closing prices of both variables. The exchange rate is defined as the number of yens per euro and each correlation coefficient is calculated using 48 weekly observations.

Suo (2002) to analyze the existence of model risk in the valuation of some exotic options. In this sense, I assume that the implied volatility surfaces of the Nikkei 225 index and the euro-yen exchange rate are generated by the local volatility model. I then calibrate the parameters of this model fitting the model to market data. For now, let assume that the instantaneous correlation between the exchange rate and the stock is known. I then apply this correlation and the implied volatilities generated by the local volatility model to price European quanto derivatives using the standard quanto adjustment under the Black-Scholes (1973) model.

Note that if we use the Black-Scholes (1973) model to price vanilla options on the Nikkei 225 index and on the euro-yen exchange rate using a volatility equal to the implied volatility generated by the local volatility model for each strike price and maturity, then we will obtain the same prices as under the local volatility model. Analogously, if we use these volatilities and the same instantaneous correlation to price quanto derivatives under the local volatility model and under the assumptions of the standard quanto adjustment, both models should yield the same price in absence of model risk. But, as we will see now, the pricing discrepancies can be very huge, above all for long-term maturities, and these discrepancies depend crucially on the values of the model parameters, such as the instantaneous correlation.

3.1 The calibration to market data

The implied volatility surfaces corresponding to the Nikkei 225 index and to the euro-yen exchange rate, as well as the dividend yield and interest rates are obtained from Bloomberg. I consider market data corresponding to August 10, 2010 and use the local volatility model to provide a close fit as possible to the market implied volatility surfaces. To this end, I use the approach introduced by Marabel (2010) to calculate the local volatility. This methodology consists of smoothing the implied volatility through a flexible parametric function, which is consistent with the no-arbitrage conditions developed by Lee (2004) for the asymptotic behavior of the implied volatility at extreme strikes. The local volatility function is then calculated analytically. This approach allows obtaining smooth and stable local volatility surfaces while capturing the prices of vanilla options quite accurately.

Regarding the implied volatility surface corresponding to the Nikkei 225 index, the data consist of nine values of moneyness[6], ranging from 75% to 125% and eleven listed maturities, from October 2010 to June 2014. Although in the equity option market the implied volatil-

[6]The moneyness is defined as $\frac{K}{S}$, where K is the strike and S is the price of the underlying asset.

ity is expressed in terms of the strike price, in the exchange rate option market the implied volatility is expressed in terms of the option delta for a given maturity. In particular, the strike corresponding to the at-the-money volatility for each given expiry is chosen so that a put and a call have the same delta with opposite signs. The data associated with the implied volatility surface corresponding to the euro-yen exchange rate consist of eight expires, ranging from two months to five years, and, for each given expiry, the at-the-money implied volatility, as well as the volatilities corresponding to the call (put) whose delta is 25% (-25%) and 10% (-10%).

Note that, given this information, it is easy to obtain the strikes corresponding to each volatility point. In particular, let σ_Z^{atm} denote the at-the-money implied volatility corresponding to the expiry $t = T$. In this case, taking into account the delta of the call and of the put under the Black-Scholes (1973) model, the at-the-money strike, denoted K_Z^{atm}, must verify:

$$\Phi(h) = \Phi(-h)$$

where

$$h = \frac{1}{\sigma_Z^{atm}\sqrt{T}}\left[\ln\left(\frac{Z_0 e^{(r_Y - r_X)T}}{K_Z^{atm}}\right) + \frac{T\left(\sigma_Z^{atm}\right)^2}{2}\right]$$

Hence, the at-the-money strike is given by:

$$K_Z^{atm} = Z_0 e^{(r_Y - r_X)T + \frac{T}{2}\left(\sigma_Z^{atm}\right)^2}$$

The other strike prices can be obtained in a similar fashion. In this sense, let $\sigma_Z^{\Delta\gamma}$ denote the implied volatility of a European call with delta equal to $\Delta\gamma$ and expiry equal to $t = T$. The corresponding strike price $K_Z^{\Delta\gamma}$, must satisfy:

$$e^{-Tr_X}\Phi\left[\frac{1}{\sigma_Z^{\Delta\gamma}\sqrt{T}}\left(\ln\left(\frac{Z_0 e^{(r_Y - r_X)T}}{K_Z^{\Delta\gamma}}\right) + \frac{T\left(\sigma_Z^{\Delta\gamma}\right)^2}{2}\right)\right] = \Delta\gamma$$

Therefore, we have:

$$K_Z^{\Delta\gamma} = Z_0 e^{(r_Y - r_X)T + \frac{T}{2}\left(\sigma_Z^{\Delta\gamma}\right)^2 - \beta\sigma_Z^{\Delta\gamma}\sqrt{T}}$$

with $\beta = \Phi^{-1}\left(\Delta\gamma e^{Tr_X}\right)$, where $\Phi^{-1}(.)$ is the inverse standard normal cumulative distribution function.

Panel A of table 1 reports the root mean squared error (RMSE) corresponding to the difference between the Nikkei 225 index market implied volatility and the implied volatility generated by the local volatility model. The table provides information about the total RMSE and the

RMSE associated with at-the-money options, out-of-the-money puts and out-of-the-money calls. Analogously, panel B shows the estimation errors corresponding to the euro-yen exchange rate. The results show that the local volatility model provides a quite accurately fit to the implied volatility surface of the equity index and the exchange rate.

Tables 2 and 3 show, respectively, the implied volatility surfaces generated by the local volatility model corresponding to the Nikkei 225 index and the euro-yen exchange rate. On the other hand, figures 2 and 3 show the local volatility surfaces associated with these implied volatilities for both assets. In both cases, the resulting local volatility surface is quite smooth and it has the properties that we would expect from a reasonable local volatility surface. In particular, the local volatility surface corresponding to the Nikkei 225 index has a pronounced skew for near-term options that decays for longer-term options. A similar pattern is found by Derman et al. (1996a), Derman et al. (1996b), as well as by Brown and Randall (1999) for the local volatility surface corresponding to the Standard and Poor's 500 index. Note that the local volatility is always positive and, therefore, the calibrated local volatility model leads to an arbitrage-free implied volatility surface.

3.2 Pricing tests

To compare the pricing performance of the standard quanto adjustment under the Black-Scholes (1973) model assumptions with the quanto adjustment under the local volatility model, I use the same value corresponding to the instantaneous correlation for both models, as well as the implied volatilities of tables 2 and 3, generated by the local volatility model, to price European quanto derivatives. If the instantaneous correlation ρ_{SZ} equals zero, there is no quanto adjustment. Hence, the relevant cases are those in which ρ_{SZ} is different from zero. Initially, I assume that ρ_{SZ} is equal to 0.75, which is a level that is consistent with the realized correlation observed using historical data.

I consider the pricing of European quanto euro calls and puts on the Nikkei 225 index, as well as the pricing of quanto euro forward contracts on this equity index.

Since the quanto adjustment has to do with the forward of the underlying asset, the potential discrepancies between different models are expected to be greater for long-term options. In this sense, I consider the pricing of quanto derivatives with expires equal to two and three years.

Given that in the local volatility framework the drift of the underlying asset price depends on the local volatility corresponding to the exchange rate, the pricing of quanto derivatives within the local volatility model has to be accomplished taking into account the joint dynamics of the equity index and the exchange rate. In this sense, applying Ito's lemma to equations (4) and (5) and integrating the result over the interval $[t, T]$ gives:

$$\frac{Z(T)}{Z(t)} = e^{(r_Y - r_X)(T-t) + \int_t^T \frac{1}{2}\sigma_Z^2(u,Z)du + \int_t^T \sigma_Z(u,Z)dW_Z^{Q_X}(u)} \tag{9}$$

$$\frac{S(T)}{S(t)} = e^{(r_Y - q)(T-t) + \int_t^T \left[\rho_{SZ}\sigma_Z(u,Z)\sigma_S(u,S) - \frac{1}{2}\sigma_S^2(u,S)\right]du + \int_t^T \sigma_S(u,S)dW_S^{Q_X}(u)} \tag{10}$$

with

$$dW_S^{Q_X}(t)\, dW_Z^{Q_X}(t) = \rho_{SZ}dt$$

I use Monte Carlo simulations with weekly time steps and 100.000 trials to estimate the prices of the quanto derivatives for the local volatility model and I apply the antithetic variable technique described in Boyle (1977) to reduce the variance of the estimates. To this end, let $[0 = t_0 < t_1 < ... < t_M = T]$ be a partition of a time interval into M equal segments of length Δt so that $t_j = j\frac{T}{M}$ for each $j = 0, 1, ..., M$. The discretization corresponding to equations (9) and (10) is given by:

$$\begin{aligned} Z(t_{j+1}) &= Z(t_j)\, e^{\left[r_Y - r_X + \frac{1}{2}\sigma_Z^2\left(t_j, Z_{t_j}\right)\right]\Delta t + \sigma_Z\left(t_j, Z_{t_j}\right)\sqrt{\Delta t}\varepsilon_{t_{j+1}}^Z} \\ S(t_{j+1}) &= S(t_j)\, e^{\left[\lambda + \rho_{SZ}\sigma_Z\left(t_j, Z_{t_j}\right)\sigma_S\left(t_j, S_{t_j}\right) - \frac{1}{2}\sigma_S^2\left(t_j, S_{t_j}\right)\right]\Delta t + \sigma_S(t_j, St_j)\sqrt{\Delta t}\varepsilon_{t_{j+1}}^S} \end{aligned}$$

where $\lambda = r_Y - q$ and where $Z(t_j)$ and $S(t_j)$ are the prices of the exchange rate and the underlying asset at time t_j, and $\varepsilon_{t_{j+1}}^Z$ and $\varepsilon_{t_{j+1}}^S$ are random samples from two standard normal distributions with correlation ρ_{SZ}.

Let consider the time $t = 0$ price of a European quanto call with maturity $t = T$ and strike price K. The estimate of the value of the option given by the local volatility model, denoted $C_{Q_X,LV}(0, T, K)$, is:

$$C_{Q_X,LV}(0, T, K) = P_X(0, T)\frac{1}{N}\sum_{i=1}^{N}\left(\widehat{S}_{i,T} - K\right)^+ \tag{11}$$

where N is the number of sample paths used in simulation and $\widehat{S}_{i,T}$ denotes the simulated value of S_T over each sample path. Analogously, the price of the quanto forward contract on the equity index under the local volatility model, denoted $F^{Q_X}_{S,LV}(0,T)$, is given by:

$$F^{Q_X}_{S,LV}(0,T) = \frac{1}{N}\sum_{i=1}^{N}\widehat{S}_{i,T} \tag{12}$$

Given the prices of the quanto call and of the quanto forward under the local volatility model, it is straight forward to obtain the price of the corresponding put using the put-call parity.

Regarding the pricing of quanto options under the assumptions of the Black-Scholes (1973) model, I consider two different approaches. The first one consists of pricing quanto options using, for each strike price, the at-the-money implied volatility of the index associated with the expiry of the option. In this case, the at-the-money volatility of the exchange rate is also used to adjust the forward of the quantoed asset. I denote this approach the pure Black-Scholes (PBS) model. Under this approach it is possible to price quanto forward contracts, as well as European quanto calls and puts using, respectively, equations (6), (7) and (8).

Since this methodology does not account for the existence of skew in the equity index, even the prices of pure vanilla out-of-the-money and in-the-money options denominated in yen will not be properly priced. Therefore, we should not expect a good performance of this model in the pricing quanto derivatives. The second approach considered, which is more outstanding, consists of using two different values for the equity index implied volatility. In this case, the at-the-money volatility is used to adjust the forward of the quantoed underlying asset, whereas the implied volatility corresponding to the option's strike and maturity is used to calculate the option premium. I denote this approach the modified Black-Scholes (MBS) model. In this sense, let consider the price of a European quanto call with maturity $t = T$ and strike price K. The time $t = 0$ price of the option under the MBS model, denoted $C_{Q_{X,MBS}}(0,T,K)$, is given by:

$$C_{Q_{X,MBS}}(0,T,K) = P_X(0,T)\left[F^{Q_X}_S(0,T)\,\Phi(d_{K,T}) - K\Phi\left(d_{K,T} - \sigma^{K,T}_S\sqrt{T}\right)\right]$$

where $F^{Q_X}_S(0,T)$ is calculated using equation (6) with σ_S and σ_Z equal to the at-the-money implied volatilities associated with expiry T and where:

$$d_{K,T} = \frac{1}{\sigma^{K,T}_S\sqrt{T}}\left[\ln\left(\frac{F^{Q_X}_S(0,T)}{K}\right) + \frac{T\left(\sigma^{K,T}_S\right)^2}{2}\right]$$

where $\sigma^{K,T}_S$ represents the implied volatility corresponding to the strike price K and the maturity T. This approach is similar to the Practitioner Black-Scholes model described by Christoffersen and Jacobs (2004) for the pricing of pure vanilla options and it is more in line with the way in which many practitioners in the financial industry accomplish the quanto correction. Note that the PBS model and the MBS model will yield the same price for at-the-money quanto options, as well as for quanto forward contracts.

The MBS model represents an attempt to price European quanto derivatives consistently with the equity market skew. But, since it only considers the at-the-money volatility associated with the exchange rate, it does not take into account the market implied volatility smile corresponding to the exchange rate. Moreover, as equation (5) proves, the existence of a local volatility smile in the exchange rate implies that the drift corresponding to the evolution of the underlying asset price depends on the local volatility function associated with the exchange rate, which is determined by the time evolution of this variable. As we will see now, this feature can have a crucial effect in the correct valuation of quanto derivatives.

Table 4 shows the prices of European quanto euro calls and puts with maturity within two years for different strike prices calculated using the local volatility model, as well as the percentage errors[7] when the option prices are calculated using the PBS model and the MBS model[8]. The table also reports the price of the quanto forward contracts under each model, as well as the dividend yield and interest rates used. As said previously, in all cases the instantaneous correlation is equal to 0.75.

The option prices corresponding to the PBS model do not consider the index implied volatility skew. Hence, as expected, options with low strikes are mispriced whereas options with high strikes are overpriced when compared with the local volatility model. In this case, the quanto forward is similar for all the models, but, as we will see below, this is not necessary true for other parameters combinations. The comparison of the option prices obtained for the MBS model and for the local volatility model shows that, in this case, the prices of calls and puts corresponding to the MBS model are overpriced for all the strike prices considered with greater discrepancies for puts and, in particular, for out-of-the-money puts. Although the pricing differences for the MBS model are lower than the differences associated with the PBS model, they are still high

[7] Option prices are expressed as a percentage of the asset price, whereas percentage errors are calculated as $\frac{BS}{LV} - 1$, where BS is the price obtained for the corresponding Black-Scholes (1973) model and LV is the price obtained using the local volatility model.

[8] I carried out some tests to check the correct implementation of the local volatility model. In particular, the prices obtained with the local volatility model and with the MBS model are equal for $\rho_{SZ} = 0$. Moreover, when $\sigma_Z(t, Z) = \sigma_Z$ and $\sigma_S(t, S) = \sigma_S$, the prices obtained under the PBS model, the MBS model and the local volatility model coincide.

indicating the existence of model risk.

Given that the magnitude of the quanto adjustment is greater the longer the time to maturity, one would expect an increase in the pricing discrepancies for longer maturities. To check this fact, table 5 reports the prices obtained when the time to maturity is equal to three years. As expected, the pricing errors associated with the MBS model are higher in absolute value for the three years maturity. Regarding the PBS model, the pricing errors are greater for at-the-money options but are lower for some other options. The reason for this effect is that the index implied volatility skew is less steep for longer maturities and, therefore, the differences of the at-the-money implied volatility with respect to the volatilities corresponding to out-of-the-money puts and calls are lower when the time to maturity is equal to three years.

Note that, once again, the value of the quanto forward is similar for all the models considered. But this contract, as the quanto options, is affected by the equity implied volatility skew, as well as by the exchange rate implied volatility smile and its value depends crucially on the level of instantaneous correlation between the underlying asset and the exchange rate. In this sense, to investigate the impact of the correlation level on the pricing discrepancies, table 6 shows the prices corresponding to a three years maturity when the instantaneous correlation is assumed to be equal to -0.75. In this case, the difference between the quanto forward under the local volatility model and under the other two models is much higher. Recall that for $\rho_{SZ} = 0.75$ puts and calls are overpriced for the MBS model when compared with the model used to generate the data, i.e. the local volatility model. Conversely, when $\rho_{SZ} = -0.75$ the puts are mispriced whereas the calls are overpriced. On the other hand, the negative value of the instantaneous correlation increases the pricing errors associated with out-of-the-money puts and out-of-the-money calls for the PBS model compared with the previous case where the instantaneous correlation had positive sign.

From the previous results, it is possible to conclude that the usual quanto adjustment can be subject to significant pricing errors for some set of parameters values. Moreover, for a given set of parameters values, pricing errors are higher for long-term maturities. These are important result, given that quanto derivatives are quite popular in the financial industry.

4 Local volatility versus stochastic volatility

As said in the introductory section, the local volatility is a deterministic function of time and the spot price. Hence, the local volatility model does not properly accounts for the existence of volatility in the volatility. Hull and Sou (2002) did not find evidence of model risk for the local volatility model in the pricing of European style options, such as compound options. But since the quanto forward depends on the joint evolution of the instantaneous volatilities associated with the underlying asset and the exchange rate, this section analyzes the importance of the existence of stochastic volatility in the evolution of the underlying asset, as well as in the evolution of the exchange rate for a correct valuation of European quanto derivatives.

To this end, I consider a multivariate version of the Heston (1993) stochastic volatility model. The main reason for this choice is that this model is one of the most popular models within the class of stochastic volatility models, since under its assumptions it is possible to obtain quasi-closed form solutions for European options. This fact allows us calibrating the model parameters to the market prices of European options.

Given the well known ability of the local volatility model to capture quite different surfaces of implied volatilities more precisely than other approaches, such as stochastic volatility models, in this section I follow Hull and Suo (2002) and I assume that market prices are governed by the multivariate stochastic volatility model. I then determine the model parameters fitting it to representative market data. Finally, I calibrate the local volatility model to the implied volatilities surfaces generated by the Heston (1993) model for the underlying asset and for the exchange rate so that both models yield the same prices for European standard options. Once this is accomplish, it is possible to compare the performance of both models in the pricing of quanto derivatives.

4.1 Multivariate stochastic volatility framework

I assume the following risk-neutral dynamics corresponding to the evolution of the asset prices and their instantaneous variances under the probability measure Q_Y:

$$\begin{bmatrix} \frac{dS(t)}{S(t)} \\ \frac{dZ(t)}{Z(t)} \\ dv_S(t) \\ dv_Z(t) \end{bmatrix} = \begin{bmatrix} (r_Y - q) \\ (r_Y - r_X) \\ \kappa_S(\theta_S - v_S(t)) \\ \kappa_Z(\theta_Z - v_Z(t)) \end{bmatrix} dt + \begin{bmatrix} \sqrt{v_S(t)} dW_S^{Q_Y}(t) \\ \sqrt{v_Z(t)} dW_Z^{Q_Y}(t) \\ \eta_S \sqrt{v_S(t)} dW_{v_S}^{Q_Y}(t) \\ \eta_Z \sqrt{v_Z(t)} dW_{v_Z}^{Q_Y}(t) \end{bmatrix} \tag{13}$$

where the correlation matrix associated with the Wiener processes is given by:

$$\Sigma = \begin{pmatrix} 1 & \rho_{SZ} & \rho_{Sv_S} & \rho_{Sv_Z} \\ \rho_{SZ} & 1 & \rho_{Zv_S} & \rho_{Zv_Z} \\ \rho_{Sv_S} & \rho_{Zv_S} & 1 & \rho_{v_Sv_Z} \\ \rho_{Sv_Z} & \rho_{Zv_Z} & \rho_{v_Sv_Z} & 1 \end{pmatrix}$$

and where the parameter θ_S represents the long-term mean corresponding to the instantaneous variance of the underlying asset, κ_S denotes the speed of mean reversion and, η_S represents the volatility of the variance. The parameters θ_Z, κ_Z and η_Z, associated with the evolution of the instantaneous variance corresponding to the exchange rate, have similar interpretation.

Since the instantaneous volatilities associated with the underlying asset and the exchange rate are adapted processes, it is possible to use the Girsanov theorem to obtain the following dynamics under the probability measure Q_X:

$$\begin{bmatrix} \frac{dS(t)}{S(t)} \\ \frac{dZ(t)}{Z(t)} \\ dv_S(t) \\ dv_Z(t) \end{bmatrix} = \begin{bmatrix} \left(r_Y - q + \rho_{SZ}\sqrt{v_S(t)}\sqrt{v_Z(t)}\right) \\ (r_Y - r_X + v_Z(t)) \\ a_S(t) \\ a_Z(t) \end{bmatrix} dt + \begin{bmatrix} \sqrt{v_S(t)}dW_S^{Q_X}(t) \\ \sqrt{v_Z(t)}dW_Z^{Q_X}(t) \\ \eta_S\sqrt{v_S(t)}dW_{v_S}^{Q_X}(t) \\ \eta_Z\sqrt{v_Z(t)}dW_{v_Z}^{Q_X}(t) \end{bmatrix} \tag{14}$$

with:

$$\begin{aligned} a_S(t) &= \kappa_S(\theta_S - v_S(t)) + \rho_{Zv_S}\eta_S\sqrt{v_S(t)}\sqrt{v_Z(t)} \\ a_Z(t) &= \kappa_Z(\theta_Z - v_Z(t)) + \rho_{Zv_Z}\eta_Z v_Z(t) \end{aligned}$$

where the correlation matrix corresponding to the Wiener processes associated with the probability measure Q_X is given by Σ.

Regarding the calibration of the parameters of this multivariate version of the Heston (1993) stochastic volatility model, we can use the well known pricing formula for European options[9] to calibrate the parameters θ_S, κ_S, η_S, ρ_{Sv_S} and $v_S(0)$ (the initial level corresponding to the instantaneous variance) to the European options corresponding to the Nikkei 225 index. Analogously, it is possible to determine the model parameters θ_Z, κ_Z, η_Z, ρ_{Sv_Z} and $v_Z(0)$, fitting the model to the market prices associated with European options for the euro-yen exchange rate.

With regard to the estimation of the instantaneous correlation between the underlying asset and the exchange rate ρ_{SZ}, I consider the realized correlation observed using historical data. For the inter-variance correlations and cross-asset-variance correlations, I follow the method proposed by Jäckel and Kahl (2009) which is also used by Jäckel (2010b). In particular, the cross correlations are determined as follows:

$$\begin{gathered} \rho_{Sv_Z} = \rho_{SZ}\rho_{Zv_Z} \\ \rho_{Zv_S} = \rho_{SZ}\rho_{Sv_S} \\ \rho_{v_S v_Z} = \rho_{SZ}\rho_{Sv_S}\rho_{Zv_Z} + \psi\sqrt{1-\rho_{Sv_S}^2}\sqrt{1-\rho_{Zv_Z}^2} \end{gathered} \tag{15}$$

[9] I use the fast Fourier transform proposed by Carr and Madan (1999) to improve the accuracy of the results.

with $-1 \leq \psi \leq 1$. The parameterization of equation (15) allows for positive semi-definite correlation matrices for a wide range of variance-variance correlations.

4.2 Market data calibration

When considering Nikkei 225 index options, model parameters are chosen to provide as close a fit as possible to the implied volatility surface corresponding to August 10, 2010 considered in section 3.1. The estimated parameters are $\theta_S = 0.0798$, $\kappa_S = 2.0663$, $\eta_S = 0.4030$, $\rho_{Sv_S} = -0.7356$ and $v_S(0) = 0.0606$. Figure 4 shows the implied volatility surface generated by this parameter set. The figure reveals the existence of negative volatility skew, which is most pronounced for near-term options. This is a common pattern of behavior that has been widely observed in other well known equity indexes such as the Standard and Poor's 500 index or the Eurostoxx 50 index. Note that the magnitude and sign of the correlation coefficient ρ_{Sv_S} is consistent with the negative implied volatility skew observed in the Nikkei 225 index data.

When considering the euro-yen exchange rate, the parameters of the model are chosen to provide as close a fit as possible to the implied volatility surface corresponding to August 10, 2010 considered in section 3.1. The estimated parameters are given by $\theta_Z = 0.0455$, $\kappa_Z = 4.1511$, $\eta_Z = 0.6000$, $\rho_{Zv_Z} = -0.2795$ and $v_Z(0) = 0.0080$. Figure 5 shows the implied volatility surface generated by these parameters. Note that the magnitude of the correlation coefficient ρ_{Zv_Z} is much lower than in the case of the equity index. As pointed out by Hull and White (1987), a low correlation level leads to the existence of volatility smile which is a common feature in the foreign exchange markets.

Panel A of table 7 reports the RMSE corresponding to the difference between the Nikkei 225 index implied volatility surface generated by the Heston (1993) model and the implied volatility calibrated using the local volatility model. The table provides information about the total RMSE and the RMSE associated with at-the-money options, out-of-the-money puts and out-of-the-money calls. Analogously, panel B shows the estimation errors corresponding to the euro-yen exchange rate implied volatility. The results show that the local volatility model provides a quite accurately fit to the implied volatility surface of the equity index and the exchange rate generated by the Heston (1993) model.

4.3 Pricing performance

Once we have calibrated the local volatility model to the implied volatilities generated by the Heston (1993) model, it is possible to compare the pricing performance of both specifications in the valuation of European quanto derivatives.

Regarding the multivariate stochastic volatility model, I use Monte Carlo simulations with weekly time steps and 100.000 trials to estimate the prices of the quanto derivatives. To this end, let $[0 = t_0 < t_1 < ... < t_M = T]$ be a partition of a time interval into M equal segments of length Δt so that $t_j = j\frac{T}{M}$ for each $j = 0, 1, ..., M$. The discretization corresponding to equation (14) is given by:

$$\begin{aligned}
S(t_{j+1}) &= S(t_j)\, e^{\left[r_Y - q + \rho_{SZ}\sqrt{v_S(t_j)}\sqrt{v_Z(t_j)} - \frac{1}{2}v_S(t_j)\right]\Delta t + \sqrt{v_S(t_j)}\sqrt{\Delta t}\varepsilon^S_{t_{j+1}}} \\
Z(t_{j+1}) &= Z(t_j)\, e^{\left[r_Y - r_X + \frac{1}{2}v_Z(t_j)\right]\Delta t + \sqrt{v_Z(t_j)}\sqrt{\Delta t}\varepsilon^Z_{t_{j+1}}} \\
v_S(t_{j+1}) &= \left[a_S(t_j) - \frac{\eta_S^2}{4}\right]\Delta t + \left[\sqrt{v_S(t_j)} + \frac{\eta_S}{2}\sqrt{\Delta t}\varepsilon^{v_S}_{t_{j+1}}\right]^2 \\
v_Z(t_{j+1}) &= \left[a_Z(t_j) - \frac{\eta_Z^2}{4}\right]\Delta t + \left[\sqrt{v_Z(t_j)} + \frac{\eta_Z}{2}\sqrt{\Delta t}\varepsilon^{v_Z}_{t_{j+1}}\right]^2
\end{aligned}$$

with:

$$\begin{aligned}
a_S(t_j) &= \kappa_S(\theta_S - v_S(t_j)) + \rho_{Zv_S}\eta_S\sqrt{v_S(t_j)}\sqrt{v_Z(t_j)} \\
a_Z(t_j) &= \kappa_Z(\theta_Z - v_Z(t_j)) + \rho_{Zv_Z}\eta_Z v_Z(t_j)
\end{aligned}$$

where $Z(t_j)$ and $S(t_j)$ are the prices of the exchange rate and the underlying asset at time t_j, whereas $v_Z(t_j)$ and $v_S(t_j)$ represent their instantaneous variances. Note that to improve the performance of the Monte Carlo simulations, I have implemented a Milstein discretization scheme for the instantaneous variances, as described in Gatheral (2006). Finally, the vector $\left(\varepsilon^S_{t_{j+1}}, \varepsilon^Z_{t_{j+1}}, \varepsilon^{v_S}_{t_{j+1}}, \varepsilon^{v_Z}_{t_{j+1}}\right)'$ represents a random sample for a multivariate standard normal distribution with correlation matrix Σ.

Given the previous discretization corresponding to the multivariate stochastic volatility model it is possible to price European quanto derivatives using the Monte Carlo estimators of equations (11) and (12). To this end, I consider that the instantaneous correlation between the equity index and the exchange rate ρ_{SZ} is given by the realized correlation observed using historical data and it is equal to 0.75. Taking into account the parameterization of the correlation matrix Σ corresponding to equation (15), the choice of ψ determines the correlation between both instantaneous variances. This correlation could be difficult to obtain from market data since there are not many liquid instruments in the market which are sensitive to this parameter. One alternative could be the estimation of this correlation using historical data. As an example, I calculated time series of realized variances for the weekly performance of the Nikkei 225 index and the euro-yen exchange rate, using 100 observations to calculate each variance point in the series, from February 2006 to December 2010. The correlation between the series of realized variances is equal to 0.96. Therefore, I will consider $\psi = 1$ as the starting point to compare the pricing performance of the stochastic volatility framework and the local volatility model. Under this specification, we have that the variance-variances correlation $\rho_{v_S v_Z}$ is equal to 0.8046.

Table 8 reports the prices of European quanto euro calls and puts with maturity within three years for different strike prices calculated using the multivariate stochastic volatility model, as well as the percentage errors when the option prices are calculated using the local volatility model. Option prices are expressed as a percentage of the asset price, whereas percentage errors are calculated as $\frac{LV}{SV} - 1$, where LV is the price obtained under the local volatility model and SV is the price obtained using the stochastic volatility specification. The table also shows the price of the quanto forward contracts under each model, as well as the dividend yield and interest rates used. As said previously, the instantaneous correlation between the underlying asset and the exchange rate is equal to 0.75 and $\psi = 1$.

The results of the table show that the prices of the different options are quite similar under both models. The main difference corresponds to the put with strike 60%. But we can conclude that in this case, where the magnitude of the variance-variance correlation $\rho_{v_S v_Z}$ is consistent with historical data, there are no significant pricing discrepancies between the local volatility model and the stochastic volatility specification. This result is in line with the findings of Hull and Suo (2002), who did not find evidence of model risk for the local volatility model in the pricing of European style options, such as compound options.

To analyze the sensitivity of European quanto derivatives with respect to the variance-variance correlation under the stochastic volatility specification, table 9 reports the prices obtained for $\psi = -1$. With this specification we have that $\rho_{v_S v_Z} = -0.4962$. The table also shows the percentage errors associated with the local volatility model. In this case, the calls are overpriced under the local volatility model when compared with the stochastic volatility model. Conversely, the puts are mispriced. In both cases the major discrepancies correspond to out-of-the-money options. These effects have to do with the fact that under this specification for the variance-variance correlation, the quanto forward is much lower under the stochastic volatility model than under the local volatility model.

These results show that the prices of European quanto derivatives under the stochastic volatility model are quite sensitive to the correlation between the instantaneous variances associated with the underlying asset price and the exchange rate. In this sense, when the value of this parameter is close to one, as it is the case when we consider historical data, there is little evidence of model risk when using the local volatility model to price European quanto euro derivatives on the Nikkei 225 equity index.

5 Conclusion

Quanto derivatives have become very popular since these products allow investors of a given country to invest directly in assets of countries with different currencies without taking any exposure to the evolution of the exchange rate. European quanto derivatives are usually priced using the well known quanto adjustment corresponding to the forward of the quantoed asset under the assumptions of the Black-Scholes (1973) model. But the equity option market is characterized by the existence of volatility skew and the exchange rate option market is characterized by the existence of volatility smile.

In this article, I present the quanto adjustment corresponding to the local volatility model that allows pricing quanto derivatives consistently with the market equity skew and the exchange rate smile. I then examine the model risk arising in the standard quanto adjustment by fitting the local volatility model to market data and then comparing the prices of European quanto derivatives with those generated by the standard quanto adjustment. The results show that the standard quanto adjustment is subject to significant pricing errors when compared with the local volatility model, leading to the existence of model risk. The reason for this result is that this adjustment does not account for the existence of smile in the exchange rate and it is not able to consider properly the equity implied volatility skew associated with the quantoed asset. Importantly, the pricing errors are higher for long-term maturities and depend on the parameters values, such as the level of correlation between the underlying asset and the exchange rate. These results have quite important implications not only for the pricing of European quanto derivatives but also for the correct valuation of quanto exotic products, such as quanto barrier options or quanto cliquet options, where the model dependency is huge.

The quanto correlation swap is one of the most commonly used products to trade correlation between the underlying asset and the exchange rate and allows obtaining the market implied correlation. The value of this swap is given by the difference between the quanto forward corresponding to the underlying asset and the standard quanto contract. Importantly, as tables 4 and 5 show, we can have combinations of market parameters that yield similar prices for the quanto forward under the local volatility model and under the standard quanto adjustment. In this case, let us consider two financial institutions, one using the local volatility model and the other using the standard quanto adjustment. Since the price corresponding to the quanto forward is almost the same under both models, both institutions will obtain roughly the same implied correlation level. But, as we can see from tables 4 and 5, the prices corresponding to European quanto options will be quite different. This example shows that it would be necessary a liquid market of actively traded European quanto options to avoid the possibility of model risk.

The local volatility is a deterministic function of time and the spot price. Hence, the local volatility model does not properly accounts for the existence of volatility in the volatility. However, the quanto forward is a function of the joint evolution of the instantaneous volatilities corresponding to the underlying asset and the exchange rate. To analyze the effect of the existence of stochastic volatility in the evolution of the underlying asset and the exchange rate for the pricing of European quanto derivatives, I also compare the pricing performance of the local volatility model with the prices corresponding to a multivariate version of the Heston (1993) stochastic volatility model. The empirical results show that the prices of European quanto derivatives under the stochastic volatility model are quite dependent on the correlation between the instantaneous variances corresponding to the underlying asset price and the exchange rate. Moreover, when this correlation is close to one there is little evidence of model risk for the local volatility model in the pricing of European quanto euro derivatives on the Nikkei 225 index. These results are in line with the findings of Hull and Sou (2002) who did not find evidence of model risk for the local volatility model in the pricing of European style options, such as compound options.

As said previously, the instantaneous correlation between the underlying asset and the exchange rate is a key parameter in the valuation of quanto derivatives. In general, multi-asset options are usually priced assuming constant instantaneous correlation between assets[10]. But

[10] See Krekel et al. (2004) for a comparison of different pricing methods for basket options.

recently some interesting papers have left the constant correlation assumption. In this sense, Branger and Muck (2009) introduce a multi-factor stochastic volatility model based on Wishart processes that can capture stochastic correlation between assets. Unfortunately, the model introduces many free parameters and, hence, the calibration of the model to real market data may result difficult. On the other hand, Marabel (2011) introduces a model of uncertain correlation to price options whose cross gamma changes sign depending on the relative evolution of the underlying assets. Regarding the pricing of quanto derivatives, an interesting extension could be to assume that the correlation level is a function of the underlying asset volatility and of the volatility of the exchange rate, given that usually when the financial markets exhibits higher volatility levels, the correlations between financial assets tend to be higher in absolute value. This possible extension of the model is left for future research.

List of tables

Table 1: Root mean squared error (RMSE) associated with the difference between the market implied volatility and the implied volatility generated by the local volatility model corresponding to August 10, 2010.

	Panel A Nikkei 225 index	Panel B Euro-yen currency
Total RMSE	0.24%	0.22%
RMSE at-the-money options	0.05%	0.25%
RMSE out-of-the-money puts	0.20%	0.23%
RMSE out-of-the-money calls	0.31%	0.18%

Table 2: Implied volatility surface corresponding to the Nikkei 225 index, generated by the local volatility model.

T/K	70%	80%	90%	95%	100%	105%	110%	120%	130%
0.25	40.79	35.22	29.17	26.41	23.98	21.91	20.20	17.71	16.12
0.5	37.18	32.38	27.77	25.73	23.92	22.35	21.01	18.92	17.46
1	33.52	29.80	26.53	25.11	23.83	22.70	21.70	20.06	18.79
1.5	31.58	28.53	25.92	24.78	23.76	22.84	22.02	20.62	19.51
2	30.38	27.76	25.54	24.57	23.69	22.90	22.18	20.95	19.95
2.5	29.57	27.24	25.27	24.42	23.64	22.93	22.29	21.17	20.24
3	28.98	26.86	25.08	24.30	23.59	22.94	22.35	21.32	20.45
3.5	28.53	26.57	24.92	24.20	23.54	22.94	22.39	21.42	20.61
4	28.18	26.34	24.80	24.12	23.51	22.94	22.42	21.50	20.73
4.5	27.90	26.16	24.70	24.06	23.47	22.93	22.44	21.56	20.82
5	27.66	26.00	24.61	24.00	23.44	22.92	22.45	21.61	20.89

K represents the strike as a percentage of the asset price.
T denotes the time to maturity, expressed in years.
Implied volatilities are expressed in percentage.

Table 3: Implied volatility surface corresponding to the euro-yen currency, generated by the local volatility model.

T/K	70%	80%	90%	95%	100%	105%	110%	120%	130%
0.25	26.75	22.86	18.10	15.93	14.30	13.47	13.49	15.56	18.77
0.5	25.80	22.20	18.28	16.55	15.19	14.33	14.03	14.90	17.02
1	24.83	21.79	18.80	17.49	16.41	15.60	15.09	14.95	15.76
1.5	24.52	21.87	19.38	18.30	17.36	16.61	16.06	15.55	15.75
2	24.51	22.15	19.96	19.01	18.17	17.48	16.92	16.26	16.15
2.5	24.66	22.50	20.53	19.66	18.89	18.23	17.69	16.96	16.68
3	24.90	22.88	21.06	20.26	19.54	18.91	18.39	17.63	17.25
3.5	25.18	23.28	21.57	20.82	20.14	19.54	19.02	18.25	17.81
4	25.48	23.67	22.06	21.34	20.69	20.11	19.61	18.84	18.36
4.5	25.79	24.06	22.52	21.83	21.20	20.65	20.16	19.39	18.88
5	26.11	24.44	22.96	22.29	21.69	21.15	20.67	19.90	19.38

K represents the strike as a percentage of the asset price.
T denotes the time to maturity, expressed in years.
Implied volatilities are expressed in percentage.

Table 4: Prices of European quanto derivatives with time to maturity equal to two years.

	Local volatility price		Percentage error PBS		Percentage error MBS	
Strike	Call	Put	Call	Put	Call	Put
60%	44.03%	1.70%	-3.19%	-66.37%	0.55%	30.99%
80%	27.41%	4.59%	-3.86%	-16.91%	1.85%	17.17%
100%	14.14%	10.84%	4.31%	8.22%	4.31%	8.22%
120%	5.74%	21.95%	33.71%	10.10%	8.02%	3.38%
140%	1.74%	37.46%	119.82%	6.31%	16.99%	1.54%
Forward	103.39%	103.39%	103.10%	103.10%	103.10%	103.10%

The prices have been calculated using the following parameters values: $r_Y = 0.41\%; r_X = 1.23\%; q = 2.11\%$ and $\rho_{SZ} = 0.75$. The derivatives prices, as well as the strikes are expressed as a percentage of the underlying asset price.

Table 5: Prices of European quanto derivatives with time to maturity equal to three years.

	Local volatility price		Percentage error PBS		Percentage error MBS	
Strike	Call	Put	Call	Put	Call	Put
60%	46.34%	2.37%	-2.73%	-51.68%	1.51%	31.16%
80%	30.52%	5.67%	-1.73%	-8.63%	3.44%	19.18%
100%	17.82%	12.09%	6.27%	9.56%	6.27%	9.56%
120%	9.00%	22.40%	28.39%	11.59%	10.88%	4.55%
140%	3.93%	36.44%	76.18%	8.32%	17.35%	1.98%
Forward	105.99%	105.99%	105.95%	105.95%	105.95%	105.95%

The prices have been calculated using the following parameters values: $r_Y = 0.49\%; r_X = 1.50\%; q = 2.02\%$ and $\rho_{SZ} = 0.75$. The derivatives prices, as well as the strikes are expressed as a percentage of the underlying asset price.

Table 6: Prices of European quanto derivatives with time to maturity equal to three years.

	Local volatility price		Percentage error PBS		Percentage error MBS	
Strike	Call	Put	Call	Put	Call	Put
60%	30.24%	6.04%	-7.99%	-52.56%	0.98%	-7.63%
80%	17.57%	12.49%	-9.08%	-18.85%	0.74%	-5.03%
100%	8.64%	22.68%	0.68%	-3.08%	0.68%	-3.08%
120%	3.51%	36.67%	31.46%	0.94%	1.85%	-1.89%
140%	1.19%	53.47%	104.20%	0.89%	4.33%	-1.32%
Forward	85.31%	85.31%	86.10%	86.10%	86.10%	86.10%

The prices have been calculated using the following parameters values: $r_Y = 0.49\%; r_X = 1.50\%; q = 2.02\%$ and $\rho_{SZ} = -0.75$. The derivatives prices, as well as the strikes are expressed as a percentage of the underlying asset price.

Table 7: Root mean squared error (RMSE) associated with the difference between the implied volatility generated by the Heston (1993) model and the implied volatility calibrated using the local volatility model corresponding to August 10, 2010.

	Panel A Nikkei 225 index	Panel B Euro-yen currency
Total RMSE	0.18%	0.24%
RMSE at-the-money options	0.14%	0.24%
RMSE out-of-the-money puts	0.14%	0.25%
RMSE out-of-the-money calls	0.20%	0.22%

Table 8: Prices of European quanto derivatives with time to maturity equal to three years calculated with a multivariate stochastic volatility model and with the local volatility model.

	Stochastic volatility price		Percentage error local volatility	
Strike	Call	Put	Call	Put
60%	46.64%	2.20%	0.64%	-7.27%
80%	31.86%	6.54%	0.28%	-5.66%
100%	20.37%	14.17%	-0.05%	-3.32%
120%	12.16%	25.08%	-0.49%	-2.07%
140%	6.75%	38.79%	-1.02%	-1.37%
Forward	106.49%	106.49%	106.97%	106.97%

The prices have been calculated using the following parameters values: $r_Y = 0.49\%$; $r_X = 1.50\%$; $q = 2.02\%$; $\rho_{SZ} = 0.75$ and $\psi = 1$. The derivatives prices, as well as the strikes are expressed as a percentage of the underlying asset price. See also notes for figures 4 and 5.

Table 9: Prices of European quanto derivatives with time to maturity equal to three years calculated with a multivariate stochastic volatility model and with the local volatility model.

	Stochastic volatility price		Percentage error local volatility	
Strike	Call	Put	Call	Put
60%	44.78%	2.47%	4.84%	-17.41%
80%	30.21%	7.02%	5.78%	-12.11%
100%	18.97%	14.90%	7.36%	-8.05%
120%	11.16%	26.21%	8.48%	-6.30%
140%	6.04%	40.21%	10.71%	-4.85%
Forward	104.25%	104.25%	106.97%	106.97%

The prices have been calculated using the following parameters values: $r_Y = 0.49\%$; $r_X = 1.50\%$; $q = 2.02\%$; $\rho_{SZ} = 0.75$ and $\psi = -1$. The derivatives prices, as well as the strikes are expressed as a percentage of the underlying asset price. See also notes for figures 4 and 5.

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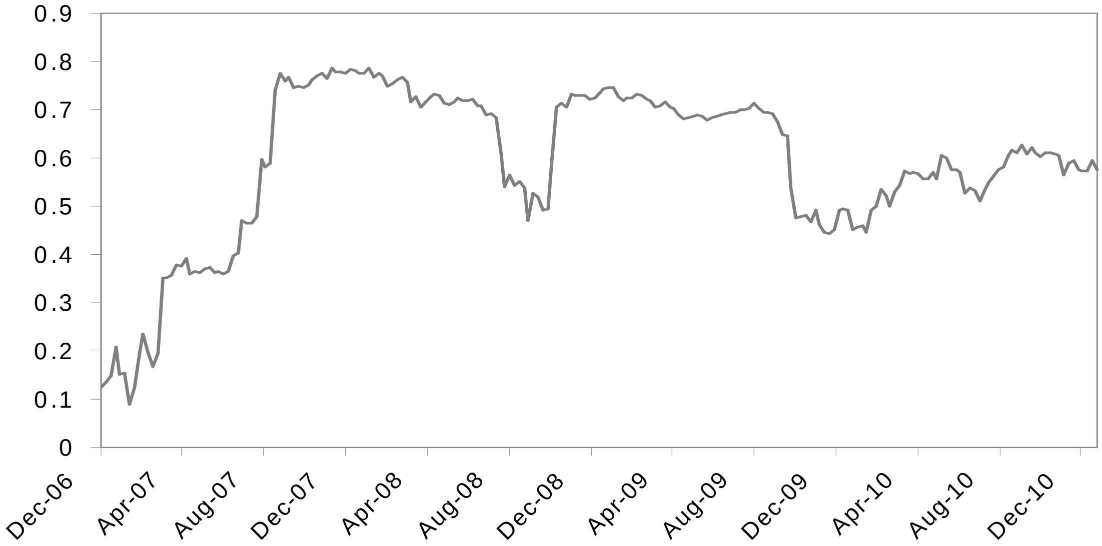

Figure 1: Realized correlation between the weekly performances of the Nikkei 225 index and the euro-yen exchange rate. The exchange rate is defined as the number of yens per euro and each correlation coefficient is calculated using 48 weekly observations.

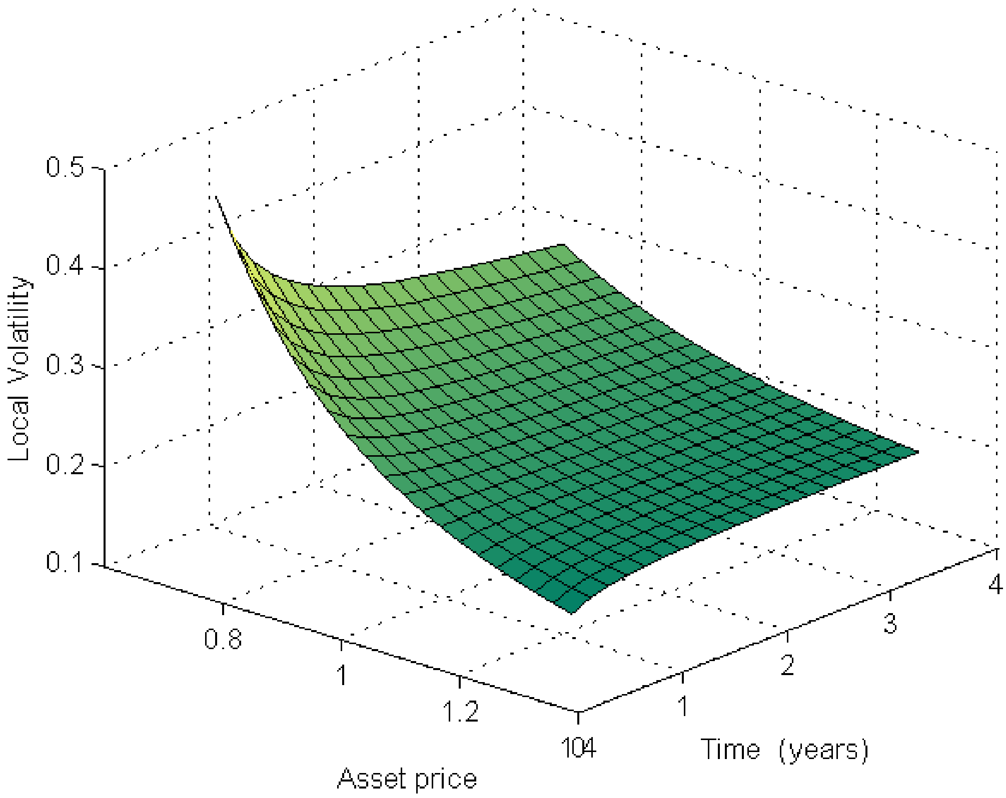

Figure 2: Local volatility surface corresponding to the Nikkei 225 equity index. Index prices are expressed as a percentage of the at-the-money strike, whereas time is expressed in years.

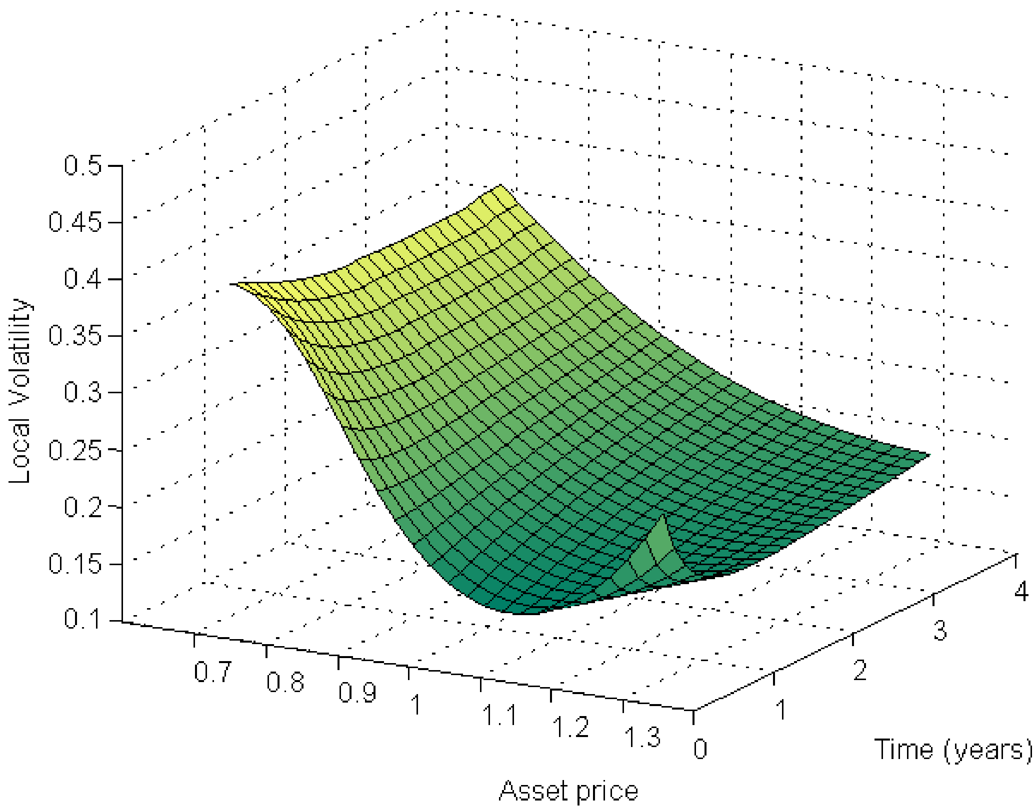

Figure 3: Local volatility surface corresponding to the euro-yen currency. Asset prices are expressed as a percentage of the at-the-money strike, whereas time is expressed in years.

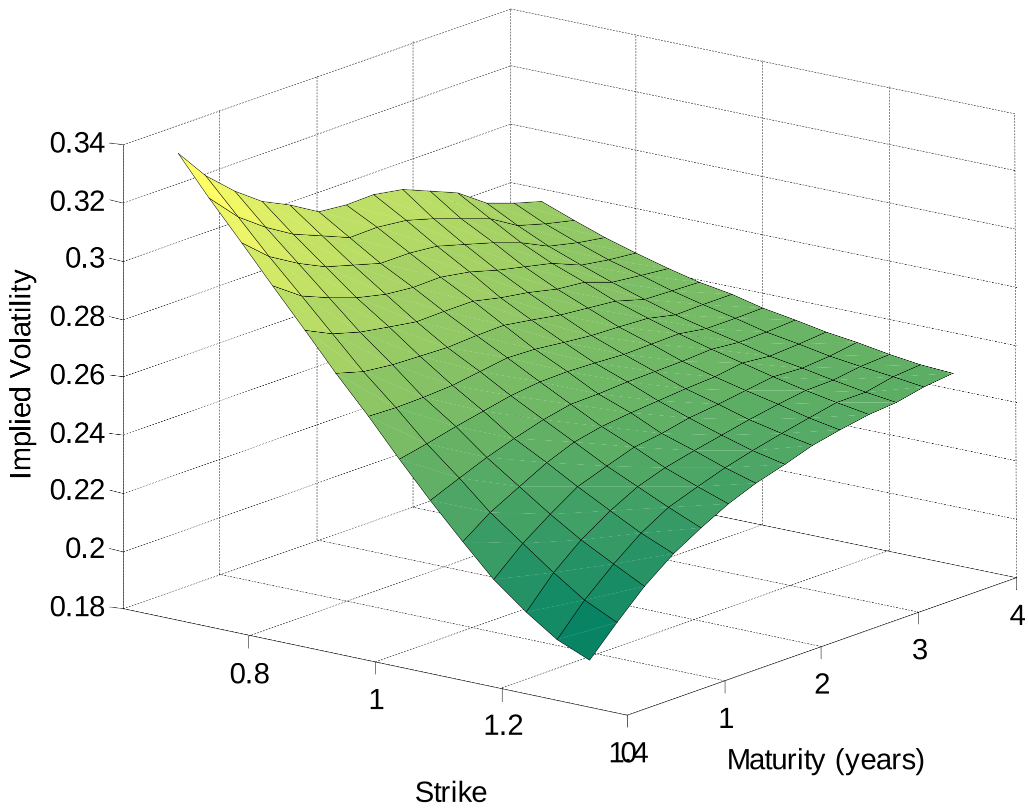

Figure 4: Implied volatility surface corresponding to the Nikkei 225 index generated by the Heston (1993) model. Strike prices are expressed as a percentage of the spot price, whereas the maturity is expressed in years. The parameters used in generating the surface are $\theta_S = 0.0798$, $\kappa_S = 2.0663$, $\eta_S = 0.4030$, $\rho_{Sv_S} = -0.7356$ and $v_S(0) = 0.0606$.

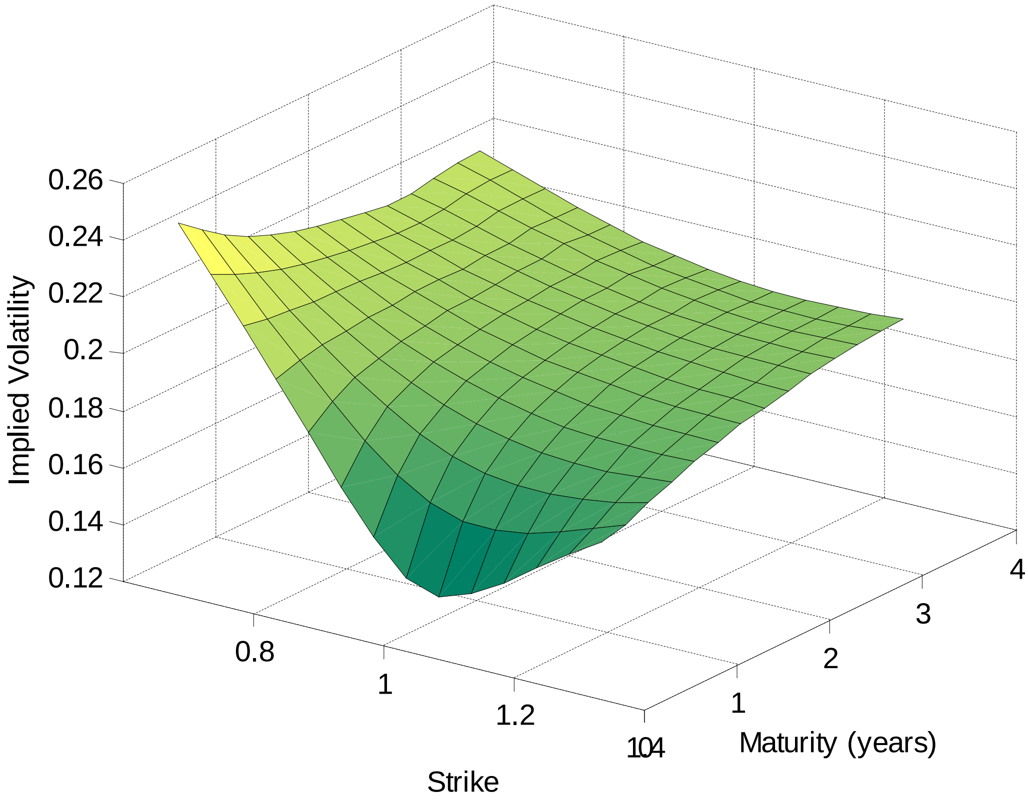

Figure 5: Implied volatility surface corresponding to the euro-yen currency generated by the Heston (1993) model. Strike prices are expressed as a percentage of the spot price, whereas the maturity is expressed in years. The parameters used in generating the surface are $\theta_Z = 0.0455$, $\kappa_Z = 4.1511$, $\eta_Z = 0.6000$, $\rho_{Zv_Z} = -0.2795$ and $v_Z(0) = 0.0080$.

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